Conformal Gravity, PT Symmetry & Indivisible Stochastic Processes

A conversation transcript — July 2026

YOU

Are you familiar with Philip Mannheim's work on conformal gravity?

CLAUDE

Yes, I'm familiar with his work. Philip Mannheim is a theoretical physicist at the University of Connecticut who has been one of the most persistent and serious advocates for conformal (Weyl) gravity as an alternative to standard Einstein gravity, particularly as a way to address the dark matter and dark energy problems without invoking unseen components.

The core idea is to replace the Einstein–Hilbert action with one built from the Weyl tensor — the action is the integral of the square of the conformal Weyl tensor, which makes the theory invariant under local conformal (scale) transformations of the metric. This local conformal symmetry is the organizing principle of the whole program. Because the action is fourth-order in derivatives of the metric rather than second-order, the field equations (the "Bach equations," with the Bach tensor set proportional to the stress–energy) are substantially different from Einstein's, and the theory has no dimensionful gravitational coupling constant put in by hand — Newton's constant is meant to emerge dynamically, tied to spontaneous breaking of the conformal symmetry.

Some of the things Mannheim is best known for in this context:

The static spherically symmetric vacuum solution (worked out with Demosthenes Kazanas) yields a potential that, beyond the usual Newtonian 1/r term, includes a linear potential term (the γr term) and a quadratic term. Mannheim has argued at length — in a long series of papers and in review articles like his "Making the Case for Conformal Gravity" — that this linear term can fit galactic rotation curves across large samples of galaxies without dark matter, using a universal contribution from the rest of the matter in the universe (a cosmological/global effect) plus a per-galaxy local term.

On cosmology, conformal gravity gives a very different picture: the conformal cosmology he develops has a naturally repulsive effective interaction and, he argues, avoids the fine-tuning of the cosmological constant problem because the vacuum energy problem is reformulated entirely — the enormous zero-point contributions are effectively decoupled from the expansion because the relevant scales are set by the same symmetry breaking that controls the matter sector.

The most serious and widely discussed objection to the program is the ghost/unitarity problem. Fourth-order derivative theories generically have a propagator that splits into pieces with opposite-sign residues, which in a canonical quantization looks like negative-norm ("ghost") states and a loss of unitarity. Mannheim (with Bender, in work on PT-symmetric quantum theory) has argued this is an artifact of assuming the Hamiltonian is

Hermitian in the conventional sense — he contends the fourth-order oscillator's Hamiltonian is not Hermitian but is PT-symmetric, and that when you construct the states in the correct (non-standard) inner product, the spectrum is real, bounded below, and the theory is unitary with no negative-norm states. This is probably his most technically distinctive contribution and remains debated.

It's worth being clear that conformal gravity is very much a minority program — most of the field regards standard general relativity plus dark matter/dark energy as far better supported, and there are ongoing disputes about whether the rotation-curve fits genuinely succeed once you look at things like galaxy clusters, gravitational lensing, and the cosmic microwave background. But Mannheim's work is careful, mathematically substantive, and he's been admirably explicit about where the theory's vulnerabilities lie.

YOU

How does his work relate at all to Jacob Barandes' work on indivisible non-Markovian processes?

CLAUDE

Honestly, the most accurate answer is that they aren't directly connected — they're separate research programs aimed at different problems, and I'm not aware of either Mannheim or Barandes substantively citing or building on the other. Mannheim's conformal gravity is a classical/gravitational field theory program (a proposed replacement for Einstein gravity), while Barandes' indivisible stochastic processes are a reconstruction and reinterpretation of quantum mechanics. So if you're looking for a documented link, I don't think there is one, and I'd be wary of anyone who claims a tidy one.

That said, there are a few genuine conceptual resonances worth drawing out, as long as we're clear these are bridges I'm constructing rather than an established relationship in the literature.

The strongest one is what I'd call the "the Hilbert-space formalism isn't fundamental" move, which both men make in different directions. Mannheim's central defense of conformal gravity — a fourth-order derivative theory that naively looks non-unitary because of ghost states — is to argue that Hermiticity is the wrong requirement. Drawing on his work with Carl Bender on PT-symmetric quantum theory, he claims the deeper principle is antilinearity (PT symmetry): once you identify the correct, non-standard inner product, the spectrum is real and bounded below and probability is conserved, so the theory is unitary after all. Barandes is likewise saying the standard apparatus — Hilbert space, unitary evolution, the Born rule — is a derived "dictionary" description rather than the bedrock, with the bedrock being a class of generalized stochastic processes. Both are relativizing the usual axioms to something they take as more basic. They disagree about what that more basic thing is (antilinear symmetry structure versus indivisible stochastic dynamics), but the philosophical maneuver rhymes.

A second, looser resonance is around memory and temporal non-locality. Barandes' whole technical hook is that quantum systems correspond to indivisible — meaning non-Markovian, non-divisible — stochastic processes: the transition matrix over a long interval doesn't factor into a product of intermediate-time transitions, so the dynamics carries "memory." Higher-derivative theories like conformal gravity have a different but not-unrelated feature: the extra time derivatives mean the instantaneous state isn't captured by position and velocity alone, and the response functions are non-local in time. It's a real thematic echo, but I'd caution against reading too much into it — the technical settings are genuinely different.

Third, and most abstractly, both frameworks are emergentist about the familiar machinery. Mannheim wants Newton's constant and particle mass scales to emerge from spontaneous breaking of conformal symmetry; Barandes wants the Hilbert space and Born rule to emerge from the stochastic correspondence. That's a shared aesthetic more than a shared mechanism.

Where they'd likely diverge sharply is on the interpretation of quantum theory itself: Barandes is offering something close to a realist, single-world, stochastic account of quantum dynamics, whereas Mannheim's PT-symmetry work is more a claim about the correct mathematical structure of quantum theory than a stance on measurement or ontology.

One caveat on my confidence: Barandes' stochastic-quantum work drew most of its attention fairly recently, and I can't verify from memory whether there's any direct engagement between the two that I'm simply not aware of.

YOU

Give your brief Barandes interpretation of the double slit experiment.

CLAUDE

Here's the gist, as his framework would handle it.

For Barandes, the particle is always in a definite configuration — it has a real position at every instant, and it goes through exactly one slit. There's an actual matter of fact about which. In that respect his account is realist and single-world: no wave physically passing through both slits, no superposed ontology, no collapse.

The interference comes entirely from the process being indivisible — non-Markovian. In an ordinary (Markovian) stochastic process you can compute the distribution at the screen by summing over the intermediate slit positions: probability of landing at x equals probability-through-top times conditional, plus probability-through-bottom times conditional. That's the law of total probability, and it's exactly the Chapman–Kolmogorov factorization of the transition matrix through the intermediate time. Barandes' claim is that a quantum process is precisely one where that factorization fails — the transition matrix from source to screen does not equal the product of source-to-slits and slits-to-screen matrices. So even though the particle really did go through one definite slit, you are not entitled to decompose the final distribution over those intermediate slit-events. The interference fringes are the visible signature of that failure of divisibility. There's nothing wavelike doing the interfering; there's a stochastic process whose intermediate marginals don't chain together the classical way.

The measurement story then falls out cleanly. In his picture, the special moments are "division events" — times at which the process momentarily behaves Markovianly, where the environment effectively registers the configuration and you're allowed to condition. Putting a which-way detector at the slits inserts a division event there. That restores divisibility across the slit-time, the law of total probability now legitimately applies, and the interference washes out into the plain classical sum of two single-slit humps. So the disappearance of fringes under which-way measurement isn't a physical collapse of anything — it's just the reintroduction of a point where the transition matrix factorizes.

The appealing part is what it dissolves: the "how does a particle go through both slits" puzzle is rejected outright — it doesn't — and the mystery is relocated to a clean, formal property of the dynamics, namely that quantum transition matrices are non-divisible except at division events. The part reasonable people push on is whether relabeling the mystery as "indivisibility" genuinely explains it or mostly repackages the Born-rule structure in stochastic language — and how the framework scales to many particles, entanglement, and field theory.

I'll flag that this is my reconstruction of how his correspondence applies to the two-slit case rather than a passage I'm quoting; the framing is faithful to the indivisible-stochastic picture, but he may carve some details differently.

YOU

My vague understanding of Philip Mannheim describes renormalization as having an exponentially decaying number of observed states and an exponentially increasing possibility of future states (that's wrong, but he talks about exponential balancing). There seemed to be an aspect to it of causality going backwards in time. Clean that up for me.

CLAUDE

Let me try to map your vague memory onto what Mannheim actually argues, because I think you're half-remembering a real and quite specific piece of his program — the unitarity story for higher-derivative (fourth-order) theories — but the "observed states decay, future states grow" framing has drifted from the actual mechanism.

The setting is this. Conformal gravity is fourth-order in derivatives, and its toy model is the Pais–Uhlenbeck oscillator. Such theories look catastrophic by the textbook reading: the energy spectrum seems unbounded below and you appear to get negative-norm "ghost" states, which would wreck unitarity. Mannheim's resolution (much of it with Carl Bender) is that the Hamiltonian in these theories is not Hermitian — it's PT-symmetric — and that the right organizing principle for quantum theory is antilinearity/time-reversal invariance rather than Hermiticity. He has a paper titled almost exactly that: antilinearity rather than Hermiticity as the guiding principle. That's where your "causality backwards in time" intuition is really coming from: the P·T that the theory is symmetric under contains T, time reversal, and T is an antilinear operation that flips the direction of time. It's not that causes literally run backward; it's that time-reversal invariance, not Hermiticity, is doing the foundational work of guaranteeing a consistent probability interpretation.

Now the "exponential balancing," cleaned up. When a Hamiltonian isn't Hermitian, its eigenstates aren't orthonormal in the naive way. You have to use a biorthogonal structure: each right-eigenvector (ket) is paired with a distinct left-eigenvector (bra), and the physical inner product is built from that pairing rather than from the ordinary dagger. When energies come in complex-conjugate pairs, one mode evolves as a decaying exponential in time and its partner as a growing exponential — and the inner product pairs the decaying mode with the growing one, so their product is time-independent and probability is conserved. That pairing is the real "balancing": not a count of observed states shrinking while future possibilities expand, but a decaying mode and a growing mode multiplied together to give a finite, constant norm. And because a state that decays going forward in time is the same object as one that grows going backward in time, the decaying/growing pair is naturally read as a forward-in-time / backward-in-time pair. That's almost certainly the germ of the "causality going backwards" image you retained.

So the corrected one-sentence version is something like: in Mannheim's treatment of these higher-derivative theories, unitarity is rescued not by Hermiticity but by an antilinear (PT / time-reversal) symmetry, under which exponentially growing and exponentially decaying modes are paired in a biorthogonal inner product so that their product is conserved — with the growing mode being the time-reverse of the decaying one.

Two honest caveats. First, I'm reconstructing which piece of his work you encountered from the fingerprints in your description — the exponentials plus the backward-time flavor point strongly at the PT-symmetry/higher-derivative unitarity material. Second, the specific quantitative phrasing you remembered doesn't correspond to a counting statement I can attribute to him, which is consistent with your own sense that it's garbled.

YOU

Is it not possible that the antilinear PT/time-reversal symmetry element could maybe predict indivisible non-Markovian processes?

CLAUDE

It's a sharp question, and there's a real kernel in it — but I'd frame the honest answer as "plausibly a necessary condition, yes; a prediction, no," and there's a specific technical obstacle that I think is where the interesting action actually is.

Start with what actually generates indivisibility in Barandes. The core of his dictionary is that the transition probabilities of the stochastic process equal the squared moduli of a unitary matrix's entries: roughly $p(j \leftarrow i; t) = |\langle j|U(t)|i\rangle|^2$, with the unitary $U(t) = e^{-iHt}$ propagating the amplitudes. Indivisibility is then a completely mechanical fact: the amplitudes compose by matrix multiplication, $U(t_2,t_0) = U(t_2,t_1)U(t_1,t_0)$, but the probabilities do not, because $|UV|^2 \neq |U|^2|V|^2$ entrywise. The discrepancy is exactly the interference cross-terms.

Now here's where Mannheim's element genuinely bites. Whether $U(t)$ evolves by pure phases or by real exponentials is precisely what the antilinear symmetry controls. Unbroken antilinear (PT/time-reversal) symmetry gives a real spectrum, so $U(t)$ is a sum of pure phases $e^{-iE_n t}$ with modulus one — genuinely unitary. The interference cross-terms in $|U|^2$ are then bounded oscillations like $\cos(\Delta E \cdot t)$, which is exactly what lets $\Gamma(t)$ be a legitimate probability-conserving stochastic matrix while still being non-divisible. If instead the antilinear symmetry is spontaneously broken, the eigenvalues come in complex-conjugate pairs, U picks up real growing/decaying exponentials $e^{\pm\Gamma t}$, and $|U|^2$ no longer normalizes to a valid stochastic map at all. So unbroken antilinear symmetry is what selects the regime that is both indivisible and a valid stochastic process.

But notice what that argument does and doesn't deliver. It shows antilinear symmetry is a necessary structural condition — it's the thing guaranteeing you even have a well-defined, probability-conserving process with bounded interference. It does not derive indivisibility from antilinearity, because the indivisibility per se comes from the complex-linear composition of amplitudes. Antilinearity tells you the phases are pure; it doesn't tell you why you square-and-sum amplitudes in the first place. So "predict" is too strong; "gatekeep the viable regime" is about right.

And here's the obstacle that I think is the genuinely interesting part. Barandes builds his correspondence on ordinary unitarity in a standard Hilbert space with the usual positive-definite inner product. Mannheim's whole point is that in the PT setting the Hamiltonian is not Hermitian in that naive inner product — unitarity is recovered only in a modified inner product (the C-operator / PT metric). To actually connect the two programs you'd have to rebuild the stochastic-quantum dictionary using Mannheim's physical inner product, and it is not obvious the resulting object is still a bona fide stochastic matrix. That's where I'd point someone who wanted to make this rigorous: it's not a slogan, it's a concrete calculation.

Two cautions so we don't build a bridge that misleads. First, there's a different link floating around: PT-symmetric systems as effective descriptions of open subsystems with balanced gain and loss, which are the usual home of "non-Markovian" dynamics. But that's open-system memory, a different animal from Barandes' indivisibility, which is intrinsic to closed unitary evolution. Second, Mannheim's decaying↔growing pairing has that "forward/backward in time" flavor, but Barandes explicitly disavows retrocausality.

Net: your instinct that antilinear time-reversal symmetry and indivisibility belong in the same conversation is more right than wrong — the symmetry is plausibly what makes the indivisible-yet-valid regime possible — but it's a necessary-condition relationship with a real unsolved step (the PT inner product) in the middle, not a prediction.

YOU

Sure

CLAUDE

(Searched the literature.)

Searched it out, and the short version is: as far as I can find, nobody has actually run the stochastic-quantum correspondence through a non-Hermitian/PT inner product, and there's no citation link between the two programs in either direction. So the bridge we were discussing is genuinely unexplored rather than quietly settled.

The specifics of what I found. Barandes' framework as it currently stands lives entirely inside standard unitarity — and, tellingly, he treats unitarity as *derived* rather than axiomatic, which is exactly the same instinct Mannheim has. But the mechanism he uses to get there is the opposite of Mannheim's. In his most recent paper (Quantum Systems as Indivisible Stochastic Processes, July 2025) he allows the raw time-evolution operator to be non-unitary as an intermediate step, and then restores unitarity by Stinespring dilation — embedding the system in a larger Hilbert space where evolution is unitary in the ordinary inner product. Mannheim's move is the reverse: keep the Hamiltonian non-Hermitian in the original space and change the inner product (the C-operator / PT metric) so that it's effectively unitary there. That contrast is the most illuminating thing the search turned up — both agree unitarity should be explained rather than assumed, but Barandes explains it by enlarging the space while keeping the metric standard, whereas Mannheim explains it by keeping the space and deforming the metric.

There's a May 2025 paper by others, "Divisible and indivisible Stochastic-Quantum dynamics," that develops the divisibility criteria further — but it explicitly stays within standard Hermitian, positive-trace-preserving maps. And Barandes mentions antiunitary/time-reversal symmetries only in the ordinary quantum-mechanical sense, not the spontaneously-broken-PT regime that is the whole point of Mannheim's program.

Net: your intuition points at a real gap in the literature, not a filled one. If someone wanted a genuinely novel result, the concrete project is exactly the calculation I flagged — redefine Barandes' transition probabilities $\Gamma = \Theta \odot \Theta^*$ using Mannheim's PT/C inner product instead of the naive one, and check whether Γ stays a legitimate stochastic matrix in the PT-unbroken phase and how it fails in the broken phase.

Sources: The Stochastic-Quantum Correspondence (arXiv:2302.10778); Quantum Systems as Indivisible Stochastic Processes (arXiv:2507.21192); Divisible and indivisible Stochastic-Quantum dynamics (arXiv:2505.08785); Antilinearity Rather than Hermiticity as a Guiding Principle for Quantum Theory (Mannheim).

YOU

Want to try it?

CLAUDE

Yeah, let's actually do it. I used the canonical PT-symmetric two-level Hamiltonian H = [[e^(iθ), s],[s, e^(−iθ)]] with $\theta = \pi/4$, and ran Barandes' $\Gamma = \Theta \odot \Theta^*$ two ways: with the naive modulus, and with Mannheim's metric.

The experiment (Python, numpy/scipy):

```
import numpy as np
from scipy.linalg import expm, sqrtm, eig
np.set_printoptions(precision=4, suppress=True)

def build_H(r, theta, s):
    return np.array([[r*np.exp(1j*theta), s],
                     [s,                 r*np.exp(-1j*theta)]], dtype=complex)

P = np.array([[0,1],[1,0]], dtype=complex)
def PT_check(H):                       # PT symmetry: P H* P == H
    return np.allclose(P @ H.conj() @ P, H)

def biorthogonal_metric(H):            # canonical metric eta = (V V^dag)^{-1}
    E, V = eig(H)
    Vinv = np.linalg.inv(V)
    eta = Vinv.conj().T @ Vinv
    return E, V, 0.5*(eta + eta.conj().T)
# ... build U = expm(-iHt); naive Gamma = |U|^2;
#     rho = sqrtm(eta); u = rho @ U @ rho^{-1}; metric Gamma = |u|^2
```

Results:

Unbroken phase (θ=π/4, s=1, spectrum {0, 1.414}):

- Naive $\Gamma = |U|^2$ has column sums [3.23, 1.30] — *not* stochastic, because U isn't unitary in the ordinary inner product.
- The biorthogonal metric η is positive-definite (eigenvalues 0.59, 3.41), satisfies $H^\dagger\eta = \eta H$, and the similarity $\rho = \eta^{1/2}$ gives a genuinely unitary u.
- Metric-corrected $\Gamma = |\eta^{1/2} U \eta^{-1/2}|^2$ = [[0.37, 0.63],[0.63, 0.37]] — exactly doubly stochastic, non-negative. A legitimate transition matrix.
- It's genuinely indivisible: $\|\Gamma(2t) - \Gamma(t)^2\| \approx 0.93$, so a real non-Markovian process, not a Markov chain in disguise.

Broken phase (θ=π/4, s=0.5, eigenvalues 0.707 ± 0.5i):

- Naive column sums [5.34, 0.54] and diverging.
- No positive-definite metric exists — every Hermitian intertwiner of $H^\dagger\eta = \eta H$ is indefinite (signatures [−1.39, 0.24] and [−1, 1]).
- $\|U(t)\|$ blows up exponentially: 1.6 → 2.0 → 3.6 → 10 → 77 as t runs 0.5 → 8. No valid stochastic matrix exists.

Here's the punchline that deflates the strong version of the conjecture. The metric-corrected amplitude $\Theta = \eta^{1/2} U \eta^{-1/2}$ is nothing other than the ordinary unitary evolution e^{-iht} of the isospectral Hermitian partner $h = \eta^{1/2} H \eta^{-1/2}$. So the "PT-corrected Barandes process" isn't a new species of indivisible process — it's the *ordinary* Barandes process you'd already get from the Hermitian Hamiltonian h. Mannheim's antilinear symmetry earns its keep as a necessary condition (it certifies that a positive metric, hence a Hermitian partner, hence a valid stochastic process, exists), but it doesn't manufacture any indivisible structure that plain unitary quantum mechanics didn't already have.

Two genuine conceptual costs surfaced. First, $\eta^{1/2}$ is not diagonal in the configuration basis — it mixes the basis states — so the rescued process lives over rotated configurations, not the original position beables; and since η depends on H, the beable basis becomes dynamics-dependent, cutting against the thing Barandes most wants. Second, the metric isn't unique, so there's a whole family of valid stochastic processes, one per metric choice.

YOU

So what did that exercise tell us? Anything insightful?

CLAUDE

The most useful thing it did was convert a "these two deep ideas must connect" intuition into a precise statement with a precise failure mode — and in doing so it located why the naive bridge can't yield new physics, and therefore where a real one would have to live.

The sharp version: the metric-corrected process is, by construction, the ordinary Barandes process of the isospectral Hermitian partner $h = \eta^{1/2} H \eta^{-1/2}$. That's not a quirk of the toy — it's an instance of Mostafazadeh's theorem: any diagonalizable non-Hermitian Hamiltonian with a real spectrum and complete biorthogonal eigenbasis is related to a genuine Hermitian Hamiltonian by exactly that similarity. So for a closed system, the entire PT/metric apparatus is similarity-equivalent to ordinary quantum mechanics. You cannot get a novel class of indivisible stochastic processes out of the metric machinery alone — it's an isomorphism back to the Hermitian case, not a generalization. The exercise rediscovers, in Barandes' stochastic language, the reason non-Hermitian QM isn't a real extension of closed-system quantum mechanics.

Second, it clarified a clean separation of labor. Mannheim's antilinear symmetry is fundamentally an *existence* statement — does a probability-conserving unitary theory exist at all? Barandes' indivisibility is a *structure* statement — given unitarity, what's the stochastic character? These compose in exactly one direction: existence gates structure (unbroken PT is precisely the condition for a valid indivisible process to exist), but existence does not generate structure. "Necessary, not productive" is the honest one-liner, now demonstrated rather than asserted.

It also closed the loop back to where we started. Your original half-memory was about Mannheim's "exponential balancing." The metric operator $\rho = \eta^{1/2}$ *is* that balancing, made mechanical: in the unbroken phase it converts the paired growing/decaying exponentials into bounded, norm-preserving phases — and bounded oscillatory phases are exactly what make $|\Theta|^2$ an interference-bearing, indivisible stochastic matrix. So "exponential balancing," "unbroken antilinear symmetry," and "a valid indivisible process exists" turned out to be three names for the same condition.

And it sharpened where a genuinely new result could still hide: exactly where the similarity-equivalence breaks — at exceptional points, where H becomes defective, η goes singular, and ρ = η^(1/2) ceases to exist, so there's no Hermitian partner to collapse back to; or in the broken phase reread as an open subsystem, which connects to Barandes' own Stinespring-dilation move. If the two "non-Markovianities" ever genuinely meet, it'll be there.

The honest caveat: it's a two-level model, so this is a clean structural picture with a robust general argument behind it, not a theorem about field theory. But the deflationary core — closed-system PT is isomorphic to Hermitian, so it gates rather than generates indivisibility — doesn't depend on the toy. The frontier question moved from "can PT predict indivisible processes?" to "does broken-PT-as-open-dynamics give indivisible processes the Hermitian case can't?" — a much better-posed question than the one we started with.

YOU

Anything else worth pursuing here?

CLAUDE

A few things, ranked honestly — most would clarify a structure rather than overturn anything, and two I'd actually bet on.

Fixed-beable / Kolmogorov-cycle characterization (do first). Our deflation came with a cost: ρ rotated the configuration basis, so the rescued beables are dynamics-dependent. The sharp question: which non-Hermitian H admit a metric that's *diagonal in the configuration basis* (preserving the original beables)? Imposing η = diag($d_1 \ldots d_n$) positive with H†η = ηH forces H_ii real and, off-diagonal, d_i/d_j = H*_ji/H_ij. Chasing consistency around loops, I strongly suspect this collapses to a Kolmogorov cycle condition (product of coupling phases around any closed loop = 1) — the detailed-balance criterion from stochastic-process theory. If right, that's a pretty result: "PT-symmetric dynamics admits a fixed-beable indivisible representation iff its couplings satisfy a Kolmogorov cycle condition." Fully tractable analytically.

Broken phase as an open subsystem (deepest). Broken-PT gain/loss systems are standardly realized as subsystems of larger Hermitian ones, and Barandes' own route to unitarity is Stinespring dilation. Dilate the broken H to a Hermitian H_big, take the ordinary Barandes process on the enlarged space, and look at the reduced process on the original configurations. Is it a well-defined sub-stochastic process, non-Markovian in the open-system sense *and* indivisible in Barandes' sense at once? This is the only place the two non-Markovianities could genuinely meet.

The exceptional point (edge probe). The deflation relies on ρ = η^(1/2) existing; right at the EP, H is defective, η singular, ρ nonexistent. Approaching the EP and watching the metric-corrected Γ degenerate would show concretely how the indivisible process dies. A short calculation targeting exactly the crack in the argument.

The Pais–Uhlenbeck oscillator (ambitious target). The genuine fourth-order, infinite-dimensional case with the real ghost — Mannheim's actual arena. Whether the correspondence even makes sense there is the real prize, and it's hard; a destination rather than a next step.

The trap to avoid. Plotting a standard non-Markovianity measure against the PT-breaking parameter likely reproduces known open-system results and quietly swaps in the *wrong* non-Markovianity (bath memory, not intrinsic indivisibility).

If you want to move, I'd run the fixed-beable / Kolmogorov calculation — tractable, directly answers the cost we uncovered, and if the hunch holds it's the most quotable thing to come out of this thread.

YOU

Can you create a comprehensive text file that I can pass off to a Claude Code session? That captures everything?

CLAUDE

(Created `PT_stochastic_handoff.md` — a self-contained briefing with background on both programs, the core question, the working script and its results, the conclusions, and a ranked actionable next-steps section with §5.1 flagged as the recommended starting calculation. Delivered alongside `pt_barandes_baseline.py`, the runnable baseline script.)

YOU

Export this conversation as a pdf.

CLAUDE

(This document.)

End of transcript. Files produced during this session: `PT_stochastic_handoff.md` (handoff briefing), `pt_barandes_baseline.py` (baseline experiment), and this PDF.